Dr. Sunilgar Laxmangar Gusai

Assistant Professor of Mathematics
Program Head, Data Science Programs
Faculty of Computer Applications, Marwadi University, Rajkot

Graph Theory → Networked Systems → Molecular & Biomolecular Graphs → Reliable Scientific AI

Ph.D. in Mathematics & GSET (Mathematical Sciences)



Contact

+91-7984345128
dr.sunilgargusai@gmail.com
Academic Portfolio
GitHub – Open Research
LinkedIn Profile
ORCID 0009-0004-0739-4812

Portfolio LinkedIn

Last updated: October 2026

Academic Profile

- Graph theorist and computational researcher working across **spectral and structural graph theory, network science, molecular and biomolecular graphs, and reliable scientific AI**. My research connects rigorous mathematical structure with reproducible computational evidence in physical, chemical, biological, and data-driven systems.
- Developed and studied **Vertex Eccentricity Labeled Energy (VELE)** and related eccentricity-sensitive spectral constructions, with current extensions spanning **power-grid vulnerability, road-network resilience, molecular representation limits, protein residue-network geometry, uncertainty under distribution shift, and formal claim–evidence consistency**.
- Maintain an active **open-research and reproducibility programme** on GitHub, with selected public repositories containing code, frozen outputs, validation workflows, figures, provenance records, and explicit claim boundaries.
- **Research Interests:** Spectral and Structural Graph Theory; Graph Energy and Eccentricity-Based Invariants; Graph Algorithms and Formal Methods; Network Science and Resilience; Molecular Graphs and QSPR/QSAR; Biomolecular / Residue-Contact Networks; Reliable Scientific Machine Learning; Conformal Prediction and Uncertainty Quantification; Distribution Shift; Explainable and Reproducible Computational Research.

Educational Qualification

Jan 2026	**Ph.D. in Mathematics**, Saurashtra University, Rajkot. Thesis title: *Discussion on Some Topics of Current Interest in Graph Theory*.
Mar 2020	**Gujarat State Eligibility Test (GSET)**, Mathematical Sciences. Qualified State Eligibility Test for Assistant Professor during M.Sc..
2018 – 2020	**M.Sc. in Mathematics**, Krantiguru Shyamji Krishna Verma Kachchh University, Kachchh, Gujarat. Percentage: **78.33%**.
2015 – 2018	**B.Sc. (Hons.) in Mathematics**, Krantiguru Shyamji Krishna Verma Kachchh University, Kachchh, Gujarat. Percentage: **78.00%**.
2015	**Higher Secondary Certificate (H.S.C.)**, Gujarat Secondary and Higher Secondary Education Board. Percentage: **73.85%**.

Educational Qualification (continued)

2013 **Secondary School Certificate (S.S.C.)**, Gujarat Secondary and Higher Secondary Education Board. Percentage: **85.33%**.

Experience

Dec 2025 – **Present** **Program Head, Data Science Programs, Marwadi University, Rajkot.**
Academic leadership for B.Sc. Data Science and M.Sc. Data Science programmes, including academic planning, curriculum coordination, course scheduling, faculty coordination, student mentoring, documentation, quality processes, and interdisciplinary integration of mathematics, data science, and computing.

Aug 2022 – **Present** **Assistant Professor of Mathematics, Faculty of Computer Applications, Marwadi University, Rajkot.**
Teaching and mentoring students across BCA, B.Sc. IT, B.Sc. Data Science, M.Sc. Data Science, M.Sc. Mathematics, and MCA. Courses include Linear Algebra, Calculus, Discrete Mathematics, Operations Research, Probability, Statistics, and Applied Mathematics, with emphasis on computational and data-oriented applications.

Aug 2023 – **Present** **Coordinator, Time Table Committee, Marwadi University, Rajkot.**
Responsible for academic scheduling, timetable coordination, documentation, and operational support across multiple programmes and semesters.

Jan 2021 – Apr 2022 **Assistant Professor (Contractual), Department of Mathematics, Saurashtra University, Rajkot.**
Delivered undergraduate and postgraduate mathematics courses, contributed to student mentoring, and supported departmental academic activities.

Academic Leadership and Administrative Contributions

2023 – **Present** **Academic Administration, Timetable Coordination and Quality Support.**
Contributing to curriculum planning, examination work, student mentoring, academic documentation, committee coordination, timetable operations, accreditation support, and quality-assurance processes.

2026 **Industrial and Academic Exposure Coordination.**
Coordinated student industrial visits and academic exposure activities including BISAG, Gandhinagar; Solvex; and Rishabh Software, with emphasis on data science, AI, Python, databases, software practice, and industry-oriented learning.

2024 – 2026 **Departmental Events and Student Engagement.**
Contributed to conference support, student activities, academic events, mentoring, and interdisciplinary coordination between mathematics, computer applications, AI, and data-science programmes.

Research Publications and Manuscripts

Published Research Articles

2026 **Reciprocal Irregularity Under Partial Edge Subdivision**.
International Journal of Advanced Multidisciplinary Research and Studies, Vol. 6(5), pp. 1336–1341, September–October 2026. Publication page: `https://www.multiresearchjournal.com/arclist/list-2026.6.5/id-7021`

Research Publications and Manuscripts (continued)

- **Edge Failure Sensitivity of Wiener and Harary Indices in Classical Network Graphs**.
  International Journal of Scientific Development and Research, Vol. 11(9), pp. b394–b402, September 2026. DOI: https://doi.org/10.56975/ijsdr.v11i9.311920
- **Eccentricity-Based Bounds for the Spectral Radius of Graph Matrices**.
  International Journal of Science and Research, Vol. 15(1), pp. 681–686, January 2026. DOI: https://doi.org/10.21275/SR26109233300

2025
- **On Vertex Eccentricity Labeled Energy of a Graph**.
  International Journal of Basic and Applied Sciences, Vol. 14(4), pp. 339–350, 2025. DOI: https://doi.org/10.14419/qtfv0860

Accepted for Publication

2026
- **Explainable Machine Learning for Real-Time Cyber Threat Detection: An Intelligent Framework for Adaptive Information Security**.
  Accepted for publication in *Computer and Decision Making (COMDEM)*, Manuscript ID: CDM_121, 30 September 2026.

Selected Manuscripts Under Review / Submitted

- **Explainable Graph Machine Learning for Leakage-Controlled Pre-Failure Edge Screening**.
  Under review, *Applied Soft Computing*.
- **Budgeted Eccentricity Reduction in Graphs: Structural Bounds and Algorithms**.
  Under review, *Discrete Applied Mathematics*.
- **Learning When to Trust Fallible Scientific Rules: Applicability-Gated Molecular Property Prediction under Chemical-Space Shift**.
  Under review, *Artificial Intelligence and Applications*.
- **Vertex Eccentricity Labeled Energy of Corona Graphs**.
  Under review, *Communications in Mathematics and Applications*.
- **Mapping the Intellectual Structure and Thematic Evolution of Nilgai (Boselaphus tragocamelus) Research: A Comprehensive Bibliometric Analysis**.
  Under review, *Annals of Arid Zone*.
- **Chronology-Aware Reliable Semiconductor Quality Prediction with Calibrated Selective Machine Learning**.
  Submitted to *Journal of Engineering and Technology for Industrial Applications (ITEGAM-JETIA)*.
- **Certified Top-k Closeness Rankings under Interval Edge-Weight Uncertainty**.
  Submitted to *IAENG International Journal of Applied Mathematics*.
- **Fault-Tolerant Test-Bank Design under Item Withdrawal: Hall-Type Certificates and Minimum-Cost Reinforcement**.
  Submitted to *IAENG International Journal of Applied Mathematics*.
- **Calibration Transfer of Conformal Prediction for Concrete Strength Across SCM Composition Regimes**.
  Submitted to *Journal of Computers, Mechanical and Management*.
- **Sharp Defect Bounds for the Sombor Index under Degree-Preserving Edge Switches**.
  Submitted to *Note di Matematica*.

Selected Current Research

- **Mechanisms of Representation Degeneracy and Generalization Limits of Classical Molecular Graph Descriptors in EGFR QSAR**.
  Revised manuscript being prepared for resubmission; public reproducibility repository maintained on GitHub.

Research Publications and Manuscripts (continued)

- **Mutation Geodesic Shielding Characterizes Residue-Network Susceptibility in Protein Stability**.
  Manuscript and reproducibility package prepared for journal submission.

Open Research and Reproducibility

Portfolio
- Maintain selected public reproducibility repositories at `https://github.com/SunilgarGusai`, supporting work across graph theory, network resilience, molecular and biomolecular graphs, and reliable scientific machine learning.

Selected
- Representative open projects include **Mutation Geodesic Shielding for Protein Stability**; **VELE Power-Grid Vulnerability Screening**; **EGFR Graph QSAR — Representation Limits**; **Applicability-Gated Molecular AI**.

Practice
- Repositories emphasize executable workflows, validation checks, frozen outputs, provenance, figure/table regeneration, negative-result retention, and explicit claim boundaries.

Research Project

2024 – Present
- **Seed Grant (Minor Research Project)**, Marwadi University, Rajkot.
  Title: *The Study on Variants of Graph Energy*.
  Role: Principal Investigator (PI).
  Funding: Rs. 80,000 (Internal Research Grant).
  Duration: April 1, 2024 – Present.
  Research focused on **spectral graph theory and graph energy measures (including VELE)**, with applications in **network analysis, structural properties of graphs, and data-driven mathematical modeling**.

Paper Presentation

Jun 2026
- **Structural Fingerprinting and Classification of Graphs Using Vertex Eccentricity Labeled Energy.**
  Presented at the *International Conference on Arts & Science, Management Studies, Medical Sciences, Engineering and Law (AMMEL–2026)*, organized in Hybrid Mode, 04–07 June 2026.

Nov–Dec 2023
- **A Study on Vertex Eccentricity Labeled Energy of a Graph.**
  Presented at the *International Conference on Graph Theory and its Applications (ICGTA 2023)*, CHRIST (Deemed to be University), Bangalore, India, 30 November – 02 December 2023.

Feb 2024
- **On Vertex Eccentricity Labeled Energy of a Graph.**
  Presented at the 2nd *International Conference on Emerging Trends & Contemporary Practices (ICETCP 2024)*, Atmiya University, Rajkot, India, 09–10 February 2024.

Participations: FDP / STTP / Workshop / Seminar / Certifications

Faculty Development Programmes (FDP)

1. Participated in the *Six-Day Online International Faculty Development Programme on "AI-Enabled Teaching Strategies: Transforming Pedagogy in Higher Education"*, organized by the Department of Visual Communication, S.A. College of Arts and Science, Chennai, Tamil Nadu, India, from 18/05/2026 to 23/05/2026.

2. Participated in the *One Week Online Faculty Development Programme on "Mathematical Intelligence: AI Perspectives and Practices"*, jointly organized by the Central University of Kashmir and Vishwakarma Institute of Technology (VIT), Pune, in association with IEEE and BEIMI Mathematics Lab, from 18/05/2026 to 22/05/2026.

3. Participated in the *Two-Week Online Faculty Development Programme on "Next-Generation Data Science and Artificial Intelligence: Industry-Driven Technologies and Solutions (DSAI-2026)"*, organized by Annant Gyan Knowledge & Skills Pvt. Ltd., from 16 March 2026 to 28 March 2026.

4. Participated in the *Five-Day Online Faculty Development Programme on "Graph-Based Machine Learning and Neural Networks: Theory, Models, and Applications"*, organized by Annant Gyan Knowledge & Skills Pvt. Ltd., from 27/01/2026 to 31/01/2026.

5. Participated in the *One Week National Level Online Hands-On FDP on "Data Analytics with Mathematics & Python"*, organized by Indian Mathematics Professors Association and Infinity Research & Development Institute, Coimbatore, from 24/11/2025 to 28/11/2025.

6. Participated in the *One Week National Level Online FDP on "Mastering Operations Research with Advanced Tools: Python, SageMath, MATLAB & LaTeX Draw"*, organized by Indian Mathematics Professors Association and Infinity Research & Development Institute, Coimbatore, from 23/06/2025 to 28/06/2025.

7. Participated in the *Five-Day Virtual Faculty Development Programme on "Interdisciplinary Research Collaborations and Innovative Teaching Methods for Mathematics"*, organized by Department of Mathematics in association with IQAC and IIC, St. Joseph's College (Arts & Science), from 17/03/2025 to 21/03/2025.

8. Participated in the *Six Days International Faculty Development Program on "Empowering Advancement in AI for Education, Healthcare and Finance"*, organized by Department of Computer Applications (MCA), Hindusthan College of Engineering and Technology, Coimbatore, from 10/03/2025 to 15/03/2025.

9. Completed the *NPTEL Faculty Development Programme (FDP) equivalent course on "Advanced Graph Theory"*, IIT Kanpur (SWAYAM), Jan–Mar 2025, with consolidated score **75% (Elite + Silver)**.

10. Participated in the *One Week Faculty Development Programme on "Statistical Applications and Data Analytics for Modern Research"*, organized by Department of Mathematics, SRM Institute of Science and Technology, Vadapalani Campus, Chennai, from 17/12/2024 to 22/12/2024.

11. Participated in the *Five Day Online Faculty Development Programme on "Research in Mathematics: Present Trends and Future Prospects"*, organized by PG and Research Department of Mathematics, Sacred Heart College (Autonomous), Tirupattur, in collaboration with Patrician College of Arts and Science, Chennai, from 18/11/2024 to 22/11/2024.

12. Participated in the *One Week International Faculty Development Program on "Essential Mathematics for Machine Learning, Deep Learning and Cyber Security"*, organized by Department of Mathematics, SRM Institute of Science and Technology, Vadapalani Campus, Chennai, from 18/10/2024 to 25/10/2024.

13. Participated in the *Five Day Faculty Development Programme on "Advances in Graph Theory and Linear Algebra (AGTLA)"*, organized by Department of Mathematics, School of Advanced Sciences, VIT–AP University, from 01/10/2024 to 05/10/2024.

14. Participated in the *Five Day National Virtual Faculty Development Programme on "Data Driven Decision Making: Applied Linear Algebra, Integrating Statistical Methods and Data Science in Research"*, organized by Department of Mathematics in association with IQAC & IIC, St. Joseph's College (Arts & Science), from 23/09/2024 to 27/09/2024.

15. Completed the *NPTEL Faculty Development Programme (FDP) equivalent course on "Introduction to Abstract and Linear Algebra"*, IIT Kharagpur (SWAYAM), Jul–Sep 2024, with consolidated score **80% (Elite + Silver)**.

16. Participated in the *One-Week National Level Faculty Development Programme (Hybrid Mode) on "Assimilation of Indian Knowledge Systems with NEP-2020: Prospect and Retrospect"*, organized by Department of Basic Sciences & Humanities and IQAC, BVRIT Hyderabad College of Engineering for Women, from 30/07/2024 to 04/08/2024.

17. Participated in the *Faculty Development Program on "Real-Life Applications of Mathematics and Actuarial Science"*, organized by Department of Mathematics and Actuarial Science, B. S. Abdur Rahman Crescent Institute of Science and Technology, from 25/07/2024 to 30/07/2024.

18. Participated in the *One Week International FDP on "Numerical Computations and Dynamical Systems (NCDS–2024)"*, organized by Department of Mathematics, Anurag University, Hyderabad, from 22/07/2024 to 26/07/2024.

19. Participated in the *One Week Faculty Development Programme on "Mathematical Foundation for AI, ML & Data Science"*, organized by Department of Mathematics, SRM Institute of Science and Technology, Vadapalani Campus, Chennai, from 11/06/2024 to 19/06/2024.

20. Participated in the *Five Day International Online FDP on "Exploring Recent Trends in Applied Mathematics and Machine Learning"*, organized by Department of Mathematics, Koneru Lakshmana Educational Foundation, Hyderabad, from 20/05/2024 to 24/05/2024.

21. Participated in the *Six Days National Level Faculty Development Programme on "AI in Learning, Teaching, Assessment and Research Practices"*, organized by Department of Mathematics, Bon Secours College for Women, Thanjavur, from 06/05/2024 to 11/05/2024.

22. Participated in the *Five Days FDP on Concept of Pure and Applied Mathematics*, jointly organized by Parul University and Gujarat Ganit Mandal, from 11/03/2024 to 15/03/2024.

23. Participated in the *Five-Day Online FDP on "Advances in Graph Theory"*, organized by Division of Mathematics, School of Advanced Sciences, Vellore Institute of Technology, Chennai, from 11/09/2023 to 15/09/2023.

Short Term Training Programme (STTP)

1. Participated in the *Online Short-Term Training Program on "Unlocking Potential: Advanced Mathematics for Engineers and Scientists"*, organized by Parul University in collaboration with Department of Mathematics, NIT Jamshedpur, from 21/10/2024 to 26/10/2024.

2. Participated in the *Seven Day Short-Term Training Programme (Virtual) on "MATLAB"*, organized by Department of Mathematics, CSSR & SRRM Degree & PG College, Kamalapuram, from 08/04/2024 to 14/04/2024.

Workshops / Training

1. Participated in the training workshop *"Transforming Higher Education with Google Gemini"*, in collaboration with STTAR, on 19/12/2025.

2. Participated in the *One-Day Workshop on "LaTeX for Scientific Documentation"*, organized by MathTech Thinking Foundation, on 29/12/2024.

3. Participated in the *One-Day Workshop on "Computing Calculus with Python"*, organized by MathTech Thinking Foundation, on 23/12/2024.
4. Participated in the *National Workshop on Computational Linear Algebra and Graph Theory (NWCLAGT 2024)*, organized by CARAMS, MAHE, Manipal, from 18/12/2024 to 20/12/2024, and earned one credit.
5. Completed the *2-Day Workshop on "Business Intelligence using Power BI"*, conducted by Skill Nation, on 04/05/2024 and 05/05/2024.
6. Participated in the *One Week National Workshop (Virtual) on "SCILAB"*, organized by Department of Mathematics, CSSR & SRRM Degree & PG College, Kamalapuram, from 29/04/2024 to 04/05/2024.

Conference / Seminar Participation

1. Participated in the *Online Foreign Guest Lecture on "Wiener Index of Graphs"* delivered by Dr. Dinesh Pandey (Wilfrid Laurier University, Canada), organized by VIT–AP University, on 29/07/2025.
2. Participated in the *National Seminar on "Recent Trends in Mathematics"*, organized by Department of Mathematics, Maharaja Krishnakumarsinhji Bhavnagar University, Bhavnagar, on 11/03/2024.
3. Participated in the *International Conference on New Age Mathematics for Data Science and Artificial Intelligence (ICNAMDA–2024)*, organized by Department of Mathematics, Rathinam College of Arts and Science, Coimbatore, on 01/03/2024.
4. Participated in the *International Conference on Applied Mathematics and Nanoscience (ICAMN–2024)*, organized by Vidya Academy of Science and Technology, Thrissur, Kerala, from 23/01/2024 to 25/01/2024.
5. Participated in the *International Conference on Recent Advancements in Graph Theory (ICRAGT–2022)*, organized by Department of Mathematics, Gujarat University, from 10/09/2022 to 11/09/2022.
6. Participated in the *18th Annual Conference of ADMA and Graph Theory Day*, organized by Department of Mathematics, Vellalar College for Women (Autonomous), Erode, Tamil Nadu, from 09/06/2022 to 11/06/2022.
7. Participated in the *National Conference on Emerging Trends in Discrete Mathematics (NCETDM–2022)*, organized by Hemchandracharya North Gujarat University, Patan, in association with Atmiya University, Rajkot, from 17/02/2022 to 19/02/2022.
8. Participated in the annual conference of *Gujarat Ganit Mandal*, organized at Ganpat University, Mehsana Urban Institute of Sciences, in December 2016.

Online Courses / Technical Certifications

1. Completed the *NPTEL Online Certification Course on "Advanced Graph Theory"*, IIT Kanpur, Jan–Mar 2025, with consolidated score **75% (Elite + Silver)**.
2. Completed the *NPTEL Online Certification Course on "Introduction to Abstract and Linear Algebra"*, IIT Kharagpur, Jul–Sep 2024, with consolidated score **80% (Elite + Silver)**.
3. Completed the Udemy certification course *"Google Sheets Bootcamp: Boost Productivity and Efficiency"*, September 2024.
4. Completed the Udemy certification course *"Python Development & Data Science: Variables and Data Types"*, September 2024.
5. Completed the Udemy certification course *"Introduction to R Programming"*, August 2024.

6. Completed the Udemy certification course *"2024 Complete Python Bootcamp from Zero to Hero in Python"*, August 2024.
7. Completed the Udemy certification course *"Python Course From Basic to Advanced"*, August 2024.
8. Completed the Udemy certification course *"Computer Vision with Python"*, August 2024.
9. Completed the Udemy certification course *"NumPy, SciPy, Matplotlib & Pandas A–Z: Machine Learning"*, August 2024.

Technical Skills and Competencies

Mathematics	Graph Theory; Spectral Graph Theory; Graph Energy; Matrix Analysis; Distance-Based Graph Invariants; Linear Algebra; Probability and Statistics; Operations Research; Mathematical Modeling and Optimization.
Research Computing	Python for mathematical and statistical computation; graph/network analysis; data preprocessing; machine-learning evaluation; uncertainty and calibration analysis; reproducible experiment workflows; checkpointed computational pipelines.
Libraries / Platforms	Working experience with Python scientific tools including NumPy, pandas, scikit-learn and NetworkX; molecular workflows using RDKit; Jupyter/Google Colab; GNU Octave/MATPOWER for validation workflows; Git and GitHub for versioned research artefacts.
Tools	LaTeX; Microsoft Office Suite; Google Workspace; GitHub Actions / CI-based validation; scientific documentation, figure/table generation, and reproducibility packaging.
Academic	Curriculum development; programme leadership; academic planning and coordination; timetable management; student mentoring; interdisciplinary teaching; examination work; documentation and quality-assurance support.

Professional Memberships

2026 – Present — **Association for Computing Machinery (ACM)**.
Professional Member (Membership No.: 0298229).
Area: Graph Theory, Network Science, Artificial Intelligence.

Achievements

2025 — **Kandala Merit Scholarship**, awarded under Deendayal Port Trust Merit Scholarship Scheme for securing First Rank in M.Sc. (Mathematics).

2020 — **Qualified GSET (Gujarat State Eligibility Test)**, Mathematical Sciences, during M.Sc. (2019), demonstrating strong foundational competence in core mathematical subjects.

2024 — **Appreciation Certificate**, Marwadi University, for contribution to Anti-Ragging Awareness Initiative.

References

Dr. Vinodray J. Kaneria
Professor and Head,
Department of Mathematics,
Saurashtra University, Rajkot
Mobile: +91-9099939481
kaneriavinodray@gmail.com

Dr. Mitesh J. Patel
Assistant Professor,
Tolani College of Arts and Science, Adipur
Mobile: +91-9429061541
miteshmaths1984@gmail.com

Statement of Research

My research programme connects **graph theory, networked systems, molecular and biomolecular graph modelling, and reliable scientific AI**. The common theme is structural reasoning: I study how mathematically defined graph structure can be converted into interpretable, testable, and reproducible evidence about complex systems.

A central strand of my work is **spectral and structural graph theory**, particularly graph energy, eccentricity-sensitive constructions, structural bounds, graph operations, and extremal behaviour. I developed and continue to study **Vertex Eccentricity Labeled Energy (VELE)**, an eccentricity-informed spectral graph invariant, together with related reconstruction, rigidity, and graph-construction questions. This theoretical work provides the mathematical foundation for several application-oriented directions.

A second strand concerns **network resilience and failure-sensitive analysis**. I investigate whether topology-derived quantities can provide useful pre-failure screening information in physical and infrastructure networks. Current work includes VELE-based screening of power-grid vulnerability validated against DC-flow behaviour, as well as leakage-controlled inspection prioritisation in flood-exposed road networks. A recurring principle in this work is to distinguish structural proxy value from physical or operational ground truth rather than treating graph metrics as automatically sufficient.

My interdisciplinary research also extends to **molecular and biomolecular graphs**. In EGFR QSAR, I study representation degeneracy and the generalization limits of classical graph descriptors under chemical-space shift. In protein stability, I investigate mutation-conditioned geodesic shielding in residue-contact networks, asking when alternative shortest-path structure can protect local network geometry from mutation. These projects combine graph-theoretic reasoning with reproducible computational experiments and explicit validity checks.

A fourth direction is **reliable scientific AI and evidence**. I work on applicability-gated scientific-rule learning, conformal prediction under composition shift, reliability under geographic transfer, explainable graph machine learning, and formal claim–evidence consistency. Across these projects, I emphasize leakage-controlled validation, uncertainty, calibration, external or grouped generalization, negative controls, transparent failure modes, and machine-checkable or reproducible evidence.

I maintain an open-research programme on GitHub so that code, frozen outputs, validation scripts, figures, and evidence boundaries remain inspectable. My long-term objective is to develop mathematically grounded methods that contribute both to graph theory and to reliable scientific computation across networks, chemistry, biology, infrastructure, and data-driven decision systems.

Statement of Teaching

My teaching philosophy is centred on strong conceptual understanding, analytical reasoning, and the ability to connect mathematics with modern computational problems. I view mathematics as a foundational language for computer science, data science, artificial intelligence, optimisation, and scientific modelling; therefore, I aim to balance theoretical rigor with meaningful applications.

I have taught undergraduate and postgraduate students across BCA, MCA, B.Sc. IT, B.Sc. Data Science, M.Sc. Data Science, and M.Sc. Mathematics. Courses include **Linear Algebra, Calculus, Discrete Mathematics, Operations Research, Probability, Statistics, and Applied Mathematics**. In classroom practice I emphasise problem solving, mathematical interpretation, and links to data analysis, algorithms, optimisation, network science, and machine-learning foundations.

As **Program Head for Data Science Programs**, I also contribute to academic planning, curriculum coordination, course scheduling, faculty coordination, student mentoring, and quality processes. These responsibilities have strengthened my focus on outcome-oriented curriculum design and on maintaining a coherent mathematical foundation across computing and data-science programmes.

I encourage students to treat computational tools as extensions of mathematical thinking rather than substitutes for it. Where appropriate, I integrate Python, spreadsheets, visualisation, numerical experimentation, and reproducible workflows into teaching so that students can verify patterns, test conjectures, and connect abstract concepts with data-driven settings.

Continuous learning is an important part of my academic practice. I regularly participate in faculty development programmes, workshops, and research activities related to mathematics, graph theory, data science, artificial intelligence, and computational methods. My overall goal is to help students develop conceptual confidence, analytical maturity, and the ability to transfer mathematical reasoning into contemporary technological and research problems.